M Esha

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.03
Sri Chaitanya Junior Kalasala	Class XII (Telangana State Board of Intermediate Education)	2023	98.20%
Kendriya Vidyalaya Sangathan	Class X (CBSE)	2021	96.60%

POSITIONS OF RESPONSIBILITY

Coordinator -- Elan & Nvision, Cultural Domain

- Directed a team of **30 volunteers** to plan and execute large-scale inter-college cultural competitions.
- Managed event **scheduling, logistics, coordination**, and on-ground **execution** across multiple competitions.
- Collaborated with core fest teams on **sponsorships, artist management, and event operations**.
- Promoted from **Deputy Coordinator (2024) to Coordinator (2025)** based on performance and leadership.

Core Member -- Books & Movies Club, IIT Hyderabad

- Maintained and organized the club's **media archive** and contributed to club activities and reading/movie sessions.

PROJECTS

Core Systems Projects

- Built an **HTTP/1.1 server from raw sockets**, implementing request parsing, header handling, static file serving, and persistent connections.
- Implemented an **LRU cache** using a custom hash map and doubly linked list with **O(1) get/put operations**.
- Added **TTL-based cache eviction and per-connection multithreading** to handle concurrent client requests.

AI-Based Email Investigation & Summarization Tool

- Reconstructed **500K+ Enron emails** into conversation threads using Message-ID/In-Reply-To metadata.
- Developed a custom **reply-chain resolver** to organize emails into coherent conversation threads.
- Built an **entity-centric search system** using spaCy NER to retrieve threads involving people and organizations.
- Applied **BART abstractive summarization** to condense complex email threads for investigative review.

Programmable Selective Noise Canceller

- Implemented real-time **noise cancellation using RLS** for full suppression of unwanted signal components.
- Designed **adaptive notch filtering** for selective suppression while retaining tonal components.

FIR & IIR Filter Implementation

- Designed fixed-point FIR and IIR filters in MATLAB using **Q(2,14) representation** for hardware implementation.
- Implemented **Direct Form-II FIR and IIR architectures in Verilog**, including pipelined and non-pipelined designs.
- Verified hardware implementations by comparing **Verilog simulation waveforms with MATLAB reference outputs**.

TECHNICAL SKILLS

Programming: C, C++, Python, MATLAB, Verilog

Software & Development: GitHub, VS Code, Linux/Ubuntu

Digital Design: RTL Design, Verilog, Digital Systems

Tools: MATLAB, LTSpice, Arduino IDE, LaTeX

RELEVANT COURSES

Computer Science: Data Structures & Applications, Introduction to Programming, Introduction to Artificial Intelligence, Machine Learning

Digital & Signal Processing: Digital Systems, RTL Design & Verification, Digital Signal Processing

Mathematics: Matrix Theory, Probability & Random Processes, Introduction to Statistics

CERTIFICATIONS AND WORKSHOPS

SPARC International Workshop on Semiconductor Manufacturing & Nanofabrication -- Phase II

Hands-on training in semiconductor fabrication and nanofabrication processes through an international workshop conducted in collaboration with institutes from Taiwan and the US.

EXTRACURRICULAR

- **Inter IIT Sports Meet** participant -- IIT Kanpur, 2024
- **Active NSO Basketball Player**
- Participant in campus sports competitions